

HW0

1.

1(a) $8 \pmod{3} = 2$

1(b) $6 \pmod{4} = 2$

1(c) $1203912 \pmod{10} = 2$

2.

2(a) $-8 \pmod{10} = 2$

2(b) $-1 \pmod{14} = 13$

2(c) $-30 \pmod{15} = 0$

3.

3(a) $3 \pmod{10} = 3$

3(b) $7 \pmod{10} = -3$

3(c) $5 \pmod{10} = -5$

3(d) $-8 \pmod{10} = 2$

3(e) $-49 \pmod{10} = 1$

4.

4(a) False

4(b) True

4(c) True

4(d) True

4(e) False

4(f) True

4(g) False

4(h) True

4(i) True

4(j) False

4(k) False

5.

5(a) Shift left 3, drop overflow, fill with 0s: **101000111101000**

5(b) Shift left 3 with wraparound: **101000111101110**

6.

6(a) $475.52 = 4.7552 * 10^2$

6(b) $-0.0045 = -4.5 * 10^{-3}$

6(c) $1.45 * 10^3 + 3.22 * 10^2 = 1772 = 1.772 * 10^3$

6(d) $(7.02 * 10^3)(4.1 * 10^{-2}) = 28.782 * 10^1 = 2.8782 * 10^2$

7.

7(a) It's around $t + 40$ seconds.

7(b) Each "combination phase" is 10 s, each pipe is 0.01 s. Add times along each path from input to output. If steps can run in parallel, use the maximum of the parallel branch totals, then continue

7(c) The drink changes after the system delay. After the bean switch at time t , the output stays the old bean drink for roughly one full pipeline delay, then transitions to the new bean drink. The dairy switch at $t + 15$ shows up later as well. I could also get a "mixed era" output where the coffee base reflects the old bean setting but the dairy reflects the new dairy setting, then later both reflect the new settings.

8.

8. 2^k combinations.

9.

9. $\text{ceil}(\log_2(N))$ rounds.

10.

so:

- 0:00–0:30: 1 = blue, 2 = black, 3 = black
- 0:30–1:00: 1 = red, 2 = red, 3 = black
- 1:00–1:30: 1 = green/yellow (user changes inside), 2 = red, 3 = red
- 1:30–2:00: 1 = pink, 2 = pink, 3 = red
- 2:00–2:30: 1 = pink (then changes later), 2 = pink, 3 = pink
- 2:30–3:00: 1 = black, 2 = black, 3 = pink
- 3:00–3:30: 1 = black (then green later), 2 = black, 3 = black
- 3:30–4:00: 1 = blue (after brief green), 2 = blue, 3 = black
- 4:00–4:30: 1 = orange, 2 = blue, 3 = blue

10(b) The output is a delayed version of the input. Changes in input don't show up right away. They appear after the paint has "stepped through" the compartments, and the output updates only on the door switching rhythm

11.

Variables stored: x , y , and array $A[0..99]$.

Total = 1 (x) + 1 (y) + 100 (A) = 102 integers.

12.

Each call has parameter x and local y , so 2 integers per stack frame

`call_me(100)` creates 100 active calls before hitting the base case

Total = $100 * 2 = 200$ integers.

13.

call_me(x) returns $x + 3$, so $\text{call_me}(7) = 10$
 $z = y + \text{call_me}(x) = 5 + 10 = 15$
 $q = z + x = 15 + 7 = 22$

HW1

1.

1(a) 00110011

unsigned: 51

signed magnitude: 51

1's complement: 51

2's complement: 51

1(b) 10000000

unsigned: 128

signed magnitude: -0

1's complement: -127

2's complement: -128

1(c) 11111111

unsigned: 255

signed magnitude: -127

1's complement: -0

2's complement: -1

1(d) 10011011

unsigned: 155

signed magnitude: -27

1's complement: -100

2's complement: -101

1(e) 10001010

unsigned: 138

signed magnitude: -10

1's complement: -117

2's complement: -118

2.

2(a) -1

signed magnitude: 10000001

1's complement: 11111110

2's complement: 11111111

2(b) -15

signed magnitude: 10001111

1's complement: 11110000

2's complement: 11110001

2(c) -67

signed magnitude: 11000011

1's complement: 10111100

2's complement: 10111101

2(d) -127

signed magnitude: 11111111

1's complement: 10000000

2's complement: 10000001

3.

3(a) $x = 10, y = -13$
Result (base 10): 23
Overflow: no

3(b) $x = 117, y = 35$
Result (base 10): 82
Overflow: no

3(c) $x = 117, y = -35$
True math result: 152, but 8-bit range is -128..127
8-bit result wraps to -104
Overflow: yes

3(d) $x = -117, y = 35$
True math result: -152, not representable
8-bit result wraps to 104
Overflow: yes

3(e) $x = -117, y = 11$
Result (base 10): -128
Overflow: no

4.

4(a) two's complement integer: -1391460350

4(b) unsigned integer: 2903506946

4(c) IEEE 754 single precision float: approximately -8.1854540507e-12

5.

I'm writing each as: sign | exponent | fraction

5(a) 11010.1110
0 | 10000011 | 101011100000000000000000

5(b) -11011.10
1 | 10000011 | 101110000000000000000000

5(c) 0.000101
0 | 01111011 | 010000000000000000000000

5(d) -1.010101
1 | 01111111 | 010101000000000000000000

5(e) 10000000010
0 | 10001001 | 000000001000000000000000

6.

Representable values are {0, 1, 2, 3, 5, 7, 11, 13}. A sum overflows if its not in that set

No overflow examples :

0 + anything in the set

$$1 + 1 = 2$$

$$1 + 2 = 3$$

$$2 + 3 = 5$$

$$2 + 5 = 7$$

$$2 + 11 = 13$$

All other pairs overflow:

$$1 + 3, 2 + 2, 3 + 3, 5 + 5, 7 + 7, 11 + 11, 13 + 13$$

$$1 + 5, 1 + 7, 1 + 11, 1 + 13$$

$$2 + 7, 2 + 13$$

$$3 + 5, 3 + 7, 3 + 11, 3 + 13$$

$$5 + 7, 5 + 11, 5 + 13$$

$$7 + 11, 7 + 13$$

$$11 + 13$$